

Response to Text (20%)

Title: Traffic congestion is a major issue in many cities in Malaysia

Problem statement: Traffic congestion in many Malaysia cities leads to longer travel times, increased air pollution, and negative impacts on daily life and economic productivity.

Source text	Main ideas from the text	Summarise/Paraphrase	Synthesis
NAME (MATRIC NO): LEE YIN SHEN (A23CS0236) Title: [COLUMNIST] Building more highways won't solve traffic congestion - reducing demand will Source reference: AWANI Columnist. (2024, November 26). <i>Building more highways won't solve traffic congestion – reducing demand will</i>. Astro AWANI. https://international.astroawani.com/malaysia-news/columnist-building-	One of the main factors that causes the traffic congestion in Malaysia is the over-reliance of private vehicle. This is due to the car-centric design of urban areas, where the city is built to accommodate cars more than people. This problem led to massive economic and time losses, specialized in the densely populated city like Kuala Lumpur and Johor. In order to solve this problem, the report mentions that merely expanding the roads by adding lanes is not a sustainable solution, as it only provides a temporary way before the congestion return. Therefore, implementation of "road diets", where reduce the lanes to	Traffic congestion in Malaysia is caused by the over-reliance on private vehicles due to car-centric urban planning. Expanding the road is not an effective way to solve this problem, because it only provide a short-term relief. The article suggests to implement “road diets”, where reduce the car lanes but improve the public transportation by enhancing first- and last-mile connectivity through lanes for bikes and scooters to encourage alternative transport.	AWANI Columnist (2024) mentions that Malaysia’s traffic congestion caused by over-reliance on private vehicles. They also mention that building more roads and lanes is an ineffective way to solve the problem. They recommend to implement “road diets” and improve accessibility for public transport. Similarly, Channel News Asia (2025) suggests that congestion charges may reduce traffic congestion. But without comprehensive public transportation systems, this charge may burden the public.

<u>more-highways-wont-solve-traffic-congestion-reducing-demand-will-498038</u>	create more space for bikes, micromobility vehicles and pedestrian paths will encourages the use of alternative transport use. This article also highlights the importance of improving the public transportation, mainly to solve the first and last mile connectivity problems. It can be solved by providing special lanes for the alternative vehicle like scooters and bicycles to enhance the accessibility to public transport.		Together, these views suggest that solving Malaysia's traffic issues need to reduce the reliance on private vehicles through both urban structural redesign and public transport accessibility improvement.
NAME (MATRIC NO): LAU YAN KAI (A23CS0098) Title: Using spike model to reduce traffic congestion and improve public transportation in Malaysia	The study examined the citizens' willingness to pay (WTP) and willingness to accept (WTA) improved public transportation in Malaysia. It focused on factors such as travel time reduction, cost reduction, and increment of parking cost for private vehicles. Using a contingent valuation survey with 442 respondents in Kajang, and the spike model to account for zero or negative responses, the study	Using a contingent valuation survey, researchers determined that reducing travel time and cost by 45% would most effectively promote the use of public transportation. The average WTP for increased parking cost was USD \$0.68, but over 26.73% of respondents rejected any improvements. The study showed strong car dependency with 59% relying on private vehicles and 94.1% owning	Based on AWANI Columnist (2024), it says that traffic congestion in Malaysia is mainly due to too many people using private transportation. This is because most cities are designed more for cars than for human beings. The author recommends that simply building more roads is not an acceptable long-term solution. Instead, the idea of 'Road Diets' is introduced, this is where car lanes are narrowed and the space is used to build bike lanes

Source reference: Almselati, A. S. I., Rahmat, R. A. O. K., Jaafar, O., & Yahia, H. A. M. (2015). Using spike model to reduce traffic congestion and improve public transportation in Malaysia. <i>Transportation Research Part D: Transport and Environment</i>, 39, 58–70. https://doi.org/10.1016/j.trd.2015.04.005	found that a 45% of reduction in travel time and cost was optimal to encourage the citizens to use public transportation with an average WTP of USD \$0.68 for parking cost increments. The result indicated that 35.09% and 34.99% were the mean WTA value respectively for travel time and cost increments, whilst the spike values showed that 26.73% and 26.51% of the respondents were unwilling to accept any improvements. Demographic data showed high private car usage (59%) and high car ownership (94.1%) with income and household size influencing preferences of using public transportation. The study also highlighted the spike model's effectiveness in providing accurate WTP or WTA estimation to provide the policymakers suggestions to improve public transportation and reduce congestion by adjusting the pricing and service quantity.	a car. Income and household size affected the preference of using public transportation. By using the spike model, the study provided the policymakers with evidence to make strategy to boost the usage of public transportation and reduce traffic congestion by increasing parking cost and enhancing quality of services.	or scooter paths, which can help individuals access buses and trains more easily. Other than this, a study by Almselati et al. (2015) found that Malaysian will be more likely to switch to public transportation if the travel time and cost are reduced. The study also shows that whilst many people are not yet ready to fully support the changes, some are willing to pay more for parking – but only if the public transport system is improved. Therefore, to reduce traffic jams, calls for both better city planning and better public transportation is required. If both are done at the same time, it makes roads more efficient for all types of transportation and making trains and buses more convenient to use, more people might leave their cars at home.
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NAME (MATRIC NO): Elijah She Yu Sheng A23CS0073 Title: Malaysia mulling traffic congestion charges, identifies Johor Bahru, KL, George Town for initial rollout Source reference: Channel News Asia. (2025, February 27). <i>Malaysia mulling traffic congestion charges, identifies Johor Bahru, KL, George Town for initial rollout.</i> https://www.channelnewsasia.com/asia/malaysia-traffic-congestion-charge-kl-johor-bahru-george-town-4205136	Many Malaysian cities, including Kuala Lumpur, Johor Bahru, and George Town, are facing traffic congestion problem that cause to longer travel time, increase air pollution, and economic losses. The federal government is studying the implementation of congestion charge as a potential solution, with initial targeting to these three cities. The preliminary findings suggest that this charge can reduce traffic volume significantly, with Kuala Lumpur potentially having a 20 percent drop of traffic congestion. However, the public transport usage still low. Therefore, imposing charges without first improving the public transportation infrastructure could unfairly burden the public users. The state official, especially in Johor, highlight that an efficient and integrated public transport system must be existed before introducing the congestion charge. They recommend the supporting measures	The Traffic congestion problem in major Malaysia's cities like Kuala Lumpur, Johor Bahru, and George Town has led to longer travel times, higher pollution levels, and economic losses. The government suggests to apply congestion charge to address this issue, with early studies suggesting a potential reduction in traffic, especially in Kuala Lumpur. However, the use of public transport still remaining low that lead to a concern that this charge will unfairly impact the public users if introduced prematurely. The state officials, especially in Johor, emphasize the need for an efficient and integrated public transport system, along with measures like park-and-ride facilities, to ensure the effectiveness and fairness of the policy.	Channel News Asia (2025) reports that Malaysia is considering congestion charges in major cities like Kuala Lumpur, Johor Bahru, and George Town to ease traffic congestion. However, without efficient and integrated public transportation, such charges may unfairly affect commuters. Supporting this, Almselati et al. (2015) emphasize that to encourage the shift from private vehicles to public transport, strategies like reducing travel time, lowering costs, and increasing parking fees are essential. Therefore, implementing congestion charges must go hand-in-hand with public transport improvements and pricing policies to ensure both effectiveness and fairness.
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	like park-and-ride facilities to encourage the use of public transport. Without these improvements, the congestion charge may worsen the socio-economic challenges rather than solve traffic issues.		
NAME (MATRIC NO): DAMIYA AINA BINTI BASIR ABD SHAMMAD (A23CS0220) Title: Solving Urban Traffic Congestion Through Data-Driven Planning Source reference: Cheng, Y., Watkins, S. J., & Anciaes, P. (2024). Chapter Six – What interventions are effective in reducing congestion? In <i>Advances in Transport Policy and Planning</i> (Vol. 6). Elsevier.	The ScienceDirect article emphasizes that many vehicles on the road is the main cause of traffic congestion in urban areas. Conventional traffic control systems frequently lack flexibility and are unable to react instantly to shifting traffic patterns. The research article focuses on using big data analytics to enhance traffic forecast and optimize vehicle flow to address this. Intelligent transportation systems (ITS) in a smart city framework can dynamically control traffic and increase road efficiency by combining various information sources, including GPS data, mobile applications, and roadside sensors.	To sum up, this article agree that the root cause of traffic congestion in cities is the overwhelming number of private vehicles. this article also emphasizes the role of big data analytics in managing traffic flow through real-time data from various sources like GPS and sensors. It suggests that traditional traffic systems are inadequate and must be replaced by intelligent systems within a smart city framework to dynamically control traffic. It is also present complementary solutions using data to understand and predict congestion	Channel News Asia (2025) noted that the government intends to impose congestion fees in crowded urban areas but cautioned that the program may not work in the absence of improved public transportation. According to Cheng, Watkins, and Anciaes (2024), big data also could help determine areas that require improvement prior to the implementation of such regulations.

https://doi.org/10.1016/bs.atpp.2023.11.002	The use of AI and big data simulations to forecast traffic patterns is also shown. Urban planners can use these findings to pinpoint areas of congestion and put more sensible rules into place.		
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